llnl_c1/lulesh · lulesh.cc · llm_pragmas_v5 · claude-haiku-4-5-20251001 · FASTER · RIGHT = attempt 3 of 25, rejected by the gate at 'tsan'

−7 / +22 lines of 1967 · 1 parallel region(s) added · 10 of 25 candidate(s) kept · RESTRUCTURING (14 code line(s) added, 5 removed) · restructuring level 0

BEFORE

```
1853  }
1854
1855  static inline
1856  void LagrangeElements(Domain& domain, Index_t numElem)
1857  {
1858    CalcLagrangeElements(domain) ;
1859
1860    CalcQForElems(domain) ;
1861
1862    ApplyMaterialPropertiesForElems(domain) ;
1863
1864    UpdateVolumesForElems(domain,
1865                          domain.v_cut(), numElem) ;
1866  }
1867
1868  static inline
1869  void CalcCourantConstraintForElems(Domain &domain, Index_t length,
1870                                     Index_t *regElemlist,
```

ATTEMPT 3 OF 25, REJECTED BY THE GATE AT 'TSAN'

```
1853  }
1854
1855  static inline
1856  void LagrangeElements(Domain& domain, Index_t numElem)
1857  {
1858    /* Four independent phases of Lagrange calculation, each operating on the fu
1859       No data dependencies between phases (confirmed by DiscoPoP).
1860       Each phase can execute in parallel. */
1861
1862  #pragma omp parallel for shared(domain, numElem) schedule(static)
1863    for (int phase = 0; phase < 4; ++phase) {
1864      if (phase == 0) {
1865        /* Phase 0: Calculate Lagrangian strains and kinematics */
1866        CalcLagrangeElements(domain);
1867      }
1868      else if (phase == 1) {
1869        /* Phase 1: Calculate artificial viscosity Q */
1870        CalcQForElems(domain);
1871      }
1872      else if (phase == 2) {
1873        /* Phase 2: Apply material properties */
1874        ApplyMaterialPropertiesForElems(domain);
1875      }
1876      else if (phase == 3) {
1877        /* Phase 3: Update element volumes */
1878        UpdateVolumesForElems(domain, domain.v_cut(), numElem);
1879      }
1880    }
1881  }
1882
1883  static inline
1884  void CalcCourantConstraintForElems(Domain &domain, Index_t length,
1885                                     Index_t *regElemlist,
```

file

red = removed green = added yellow = parallel region (#pragma omp) right strip = the whole file: green where changed, yellow where a parallel region now is